

model has its own responsibilities, which call for separate methods of monitoring and security for each layer.

Network monitoring is seemingly simple—but in reality, it's a very complex process. Mixing traditional network monitoring with security monitoring further complicates things from the design perspective, for network architects, network operations teams, and the systems administrators who manage it.

While there are multiple challenges in network monitoring, the most important one is the vast amount of data gathered by the monitoring tool, and the amount of time required to assimilate the information and apply intelligence to it, in order to achieve actionable decisions. For example, a simple promiscuous packet capture on a network card for barely ten minutes provides us with a few megabytes of data. Finding out which HTTP request failed, and why, is like finding a needle in a haystack. Another important consideration is to identify key areas, often called choke points. The incorrect selection of a choke point can result in erroneous data that doesn't accurately reflect the current performance scenario. This leads to incorrect capacity planning and security mapping.

One challenge worth mentioning is caused by the unprecedented growth of a network, a result of the organisation's growth due to business expansion or company mergers. The bigger the network, the tougher it is to visualise the scale of network infrastructure. This can result in performance bottlenecks as well as security vulnerabilities. Finally, failure to incorporate proper monitoring tools is also a challenge to be addressed by senior IT management staff. It has been observed that relying purely on commercial products actually limits a firm's ability to bring diversification into the network monitoring process. The use of appropriate FOSS tools and products is highly recommended wherever applicable. As an example, there is hardly any single commercial tool that can relate Layer 7 network monitoring to the underlying Layer 2 packet and help troubleshoot a performance problem.

Network security challenges

From the infrastructure scaling point of view, irrespective of the size of an organisation, network security is often a complex area to deal with. Since network infrastructure contains components like firewalls, routers, managed switches, etc, the configurations and settings for each of these components further add to the complexity. Also, when faced with the choices of multiple devices offered by many vendors, it is easy for a network architect to get distracted from considering an appropriate solution customised to the network. As the network grows, it can be more prone to vulnerabilities and loopholes, needing

tight security policies and careful designs, using cutting-edge technology devices and solutions.

From the security point of view, a new breed of viruses and spyware has emerged recently, which exploits the operating system as well as the networking device's vulnerabilities, and can take control to cause enough damage. Though there are multiple security solutions available, hackers are often one step ahead of the cyber-cops. It is often the case that an organisation is more prone to internal attacks than to attacks originating from outside the firm's network infrastructure. Preventing such attacks needs the latest techniques, such as the deployment of intrusion detection systems, unified threat management systems (UTMs), etc

Devices involved

Network monitoring is the term used for health monitoring, whereby monitoring utilities keep a watch on various networking components, to ensure their uptime and overall performance. Network security monitoring is at a level further down the networking layers, whereby utilities capture each and every traffic packet in a promiscuous mode, correlate it with known and unknown attacks and hacking styles, store the log for further 'deep-dive' analysis, and report the alerts. While there are known components such as intrusion-detection devices, UTMs and firewalls, there are multiple software solutions, both off-the-shelf or FOSS-based, which extend the functionalities of these components, and work on the extensive log data gathered by monitoring appliances and devices. We will discuss these solutions in more detail shortly.

Best practices in network security monitoring

Network security monitoring involves collecting network packet data, segregating it among all the 7 OSI layers, and applying intelligent algorithms to get answers to security-related questions. The purpose is to know in real-time what is happening on the network at a detailed level, and strengthen security by hardening the processes, devices, appliances, software policies, etc.

While there is no single list of practices that can cover all possible situations, we can still enumerate some best practices to implement and follow for any network infrastructure:

1. Perform network performance measurement before deploying the security monitoring solution. This is essential because security monitoring can have its own footprint on the network, especially if the monitoring solution is software-based, running on the servers.
2. If possible and affordable, deploy more than one anti-virus solution. Many anti-virus software solutions don't offer spyware detection, or don't do it right; hence, a combination is always helpful.
3. Deploy at least one FOSS packet-capturing software